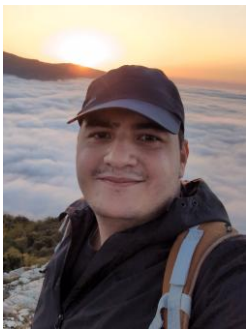
ADEL MALIK ANNABI

PhD in Mathematics — Control Theory & Applications

Inria (Team McTAO) — Université Côte d'Azur, Laboratoire J.A. Dieudonné (LJAD), Nice, France

adel.malik.annabi@proton.me · adel-malik.annabi@inria.fr

adelmalik.github.io · Google Scholar · GitHub



Research interests

Estimation · Nonlinear observer design · Optimization · Control & stabilization · Optimal control · Neuroscience

Education & professional experience

PhD in Mathematics, Inria (Team McTAO), Université Côte d'Azur (LJAD) **Oct 2022 – Jul 2026**

Thesis (defended 2 July 2026): *State and parameter estimation of neural mass and mean-field models: a control-theoretic approach*.

Supervisors: Jean-Baptiste Pomet, Dario Prandi, Ludovic Sacchelli.

Master of Mathematics & Applications — Mathematics of Modeling, Sorbonne University **2019 – 2022**

With École Polytechnique and École Nationale des Ponts et Chaussées.

Research internship at L2S — CentraleSupélec (Paris-Saclay University): *Identification of illusory state in the visual cortex*.

Supervisors: Jean-Baptiste Pomet, Dario Prandi, Ludovic Sacchelli.

Bachelor's Degree in Mathematics — Intensive Track, Sorbonne University **2017 – 2019**

Preparatory classes for the "Grandes Écoles" (CPGE), Algiers, Algeria **2015 – 2017**

Publications

Journal & conference papers

- **Identification of saturated networked systems**
  A. M. Annabi, J.-B. Pomet, D. Prandi, L. Sacchelli — *IEEE Conference on Decision and Control (CDC), Rio de Janeiro, 2025*
  DOI · HAL

- **Activity estimation via distributed measurements in an orientation sensitive neural fields model of the visual cortex**
  A. M. Annabi, J.-B. Pomet, D. Prandi, L. Sacchelli — *Mathematics of Control, Signals, and Systems (MCSS), 2025*
  DOI · HAL

- **Homogeneous observer for a low-dimensional neural fields model of cortical activity**
  A. M. Annabi, J.-B. Pomet, D. Prandi, L. Sacchelli — *IEEE Conference on Decision and Control (CDC), Milan, 2024*
  DOI · HAL

Preprints

- **Controlled Excitation for Identification of Nonlinear Systems with Saturated Outputs**
  A. M. Annabi, J.-B. Pomet, D. Prandi, L. Sacchelli — *Preprint, HAL:hal-05854609, 2026*
  Extension of thesis Chapter 8 — finite-dimensional systems
  HAL

- **Observer design for Lur'e-type systems via injection of a reconstructed nonlinear output**
  Adel Malik Annabi — *Preprint, arXiv:2606.24656 [math.OC], 2026*
  arXiv

In preparation

- **Estimation of a hidden memoryful current in neural fields**
  Adel Malik Annabi — *In preparation, 2026*

- **Saturation Excitation in Infinite Dimensions with Application to Neural Field Kernel Identification**
  A. M. Annabi, J.-B. Pomet, D. Prandi, L. Sacchelli — *In preparation, 2026*
  Extension of thesis Chapter 9 — infinite-dimensional systems

- **State and parameter estimation of triangular layered networks with saturated interconnections**
  A. M. Annabi, J.-B. Pomet, D. Prandi, L. Sacchelli — *In preparation, 2026*
  Extension of thesis Chapter 10 — two-layer networks of finite-dimensional systems

Thesis

- **State and parameter estimation of neural mass and mean-field models: a control-theoretic approach**
  Adel Malik Annabi — *PhD thesis, Université Côte d'Azur, 2026*
  Manuscript (PDF)

Conferences & talks

Talks & posters

- **IEEE Conference on Decision and Control (CDC) — Rio de Janeiro, Brazil** **Dec 2025**
  Talk — "Identification of saturated networked systems." · cdc2025.ieeecss.org
- **IEEE Conference on Decision and Control (CDC) — Milan, Italy** **Dec 2024**
  Talk — "Homogeneous observer for a low-dimensional neural fields model of cortical activity." · cdc2024.ieeecss.org
- **NeuroPSI–Chen Institute Conference — Saclay, France** **May 2024**
  Poster — "Neuroengineering for Exploring and Repairing the Brain." · neuropsi-2024.sciencesconf.org
- **Complex Days — Complex Systems Academy, Université Côte d'Azur — Nice, France** **Feb 2024**
  Poster presentation. · univ-cotedazur.eu/events/complex-days

Participation

- **NOCCA — Nonlinear and Optimal Control on the Côte d'Azur — Sophia Antipolis, France** **Sep 2026**
  nocca-2026.fr

Summer & autumn schools

- **EECI International Graduate School on Control (IGSC) — Cagliari, Italy** **Jun 2023**
  An overview on observer design methods for nonlinear systems — P. Bernard, V. Andrieu, D. Astolfi · eeci-igsc.eu
- **EITN–EBRAINS Fall School in Computational Neuroscience — Marseille, France** **Oct 2024**
  10-day course — from cellular models to whole-brain dynamics · ebrains.eu

Teaching

Teaching Assistant — Mathematics (96 h), Laboratoire J.A. Dieudonné, Université Côte d'Azur

Technical skills

Programming: Python · Julia · LaTeX · Git
Machine learning & deep learning: NumPy · SciPy · scikit-learn · PyTorch · TensorFlow/Keras